

RENSSELAER MECHATRONICS

DC Motor Speed and Direction - Step Response

Part 1: Motor Step Response

Objectives:

- Use the DRV8833 or SN754410 motor driver and obtain the motor step response
- Understand the logic needed to control the magnitude and direction of a motor
- Observe the different effects of different logic schemes to drive a motor
- Observe the effect of sampling rate on a discrete sensor
- Calculate velocity and understand unit conversions

Background Information:

A motor driver chip converts a lower power signal from the microcontroller (3.3v, 40mA) to a high-power signal (9v, 1.5A) to drive the motor. It can be thought of as a switch or relay to the driver supply voltage (9v batteries or 5v for USB in this case). Since the supply voltage is fixed this would result in a fixed motor speed. To regulate the speed a PWM signal is used to quickly switch the output on and off so that the average output voltage can be controlled.

The term motor “driver” is also commonly called “amplifier” or “chopper”. The DRV8833 driver is rated at providing 2.7-10.8v at 1.5A RMS on each channel.

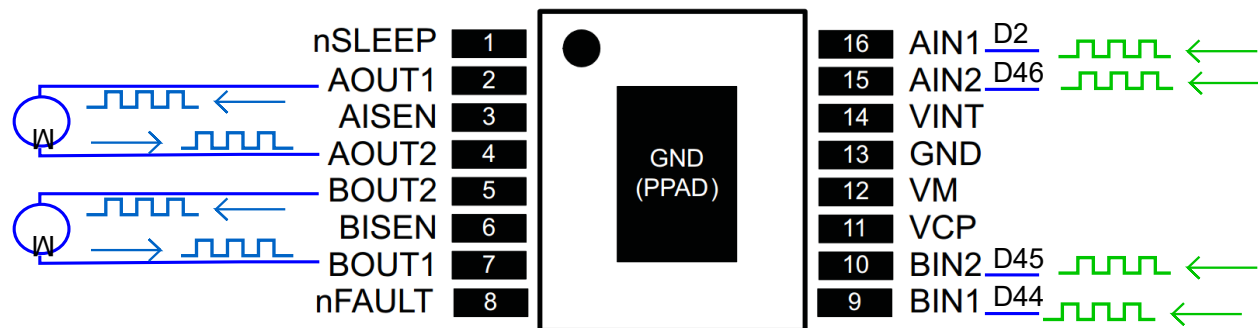


Figure 1: DRV8833 pinout/wiring diagram

The SN754410 driver is capable of providing 3.5-36v at 1A on each channel.

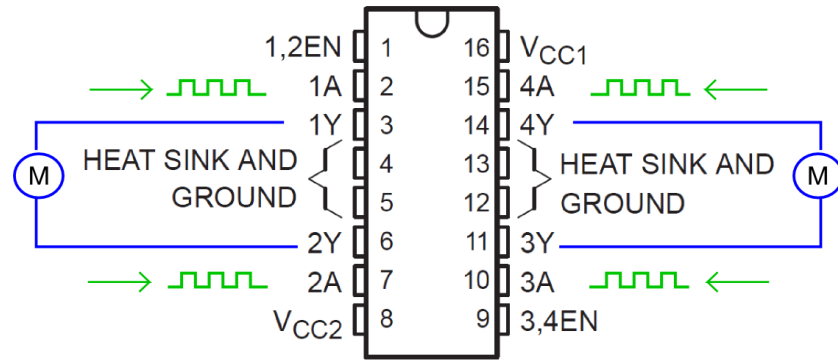
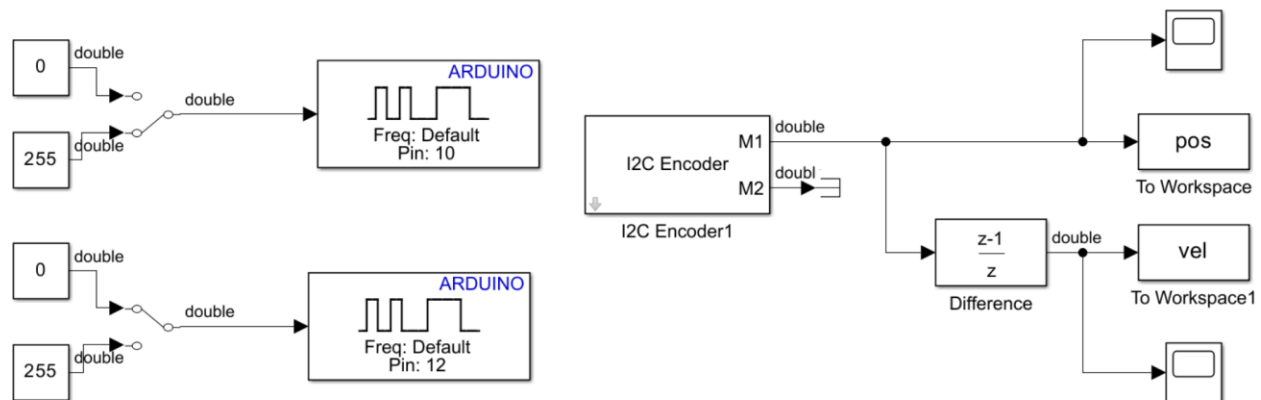


Figure 2: SN754410 pinout/wiring diagram

The green lines show the lower power PWM inputs, that get amplified then sent to the output pins to the motors.

Simulink Model – Basic Magnitude and Direction control with PWM

Build the following Simulink diagram starting from the demo file. Right click the 'I2C Motor Driver DRV8833 PWM' block and select 'uncomment' to make this block active. This diagram allows control of the PWM signal to each M1 motor terminal, and the manual switch allows each to be fully ON (full driver voltage - PWM of 255) or fully OFF (zero driver voltage - PWM of 0). The switching allows full speed operation with direction control.



- Use the following settings for the model:
 - Use a sample time of 0.03 seconds.
 - Make sure the 3 switches on the hardware are set to
 - Driver Enable: ON
 - Driver Volts: +3v
 - Power Switch: OFF
- Run the Simulink diagram with Monitor & Tune

- toggle combinations of the manual switches in the Simulink diagram to observe the results.
- While the motor is running change the value of 255 to a smaller value and observe the results.

This is the basic logic that is used to control the speed and direction of a motor using two amplified PWM signals.

- Stop the model then plot the results using the sample code below. The commented line below stores the variables 'pos' and 'vel' into the data file 'ext_dat.mat'. Copy and paste the commented out save command in the command line to save the data. **You do not want this save command uncommented in an m-file since it might cause you to accidentally overwrite previously stored data file when you run the m-file.**

```
%save ext_dat pos vel % save the data
load ext_dat % load the data
figure, plot(pos)
figure, plot(vel), hold on
```

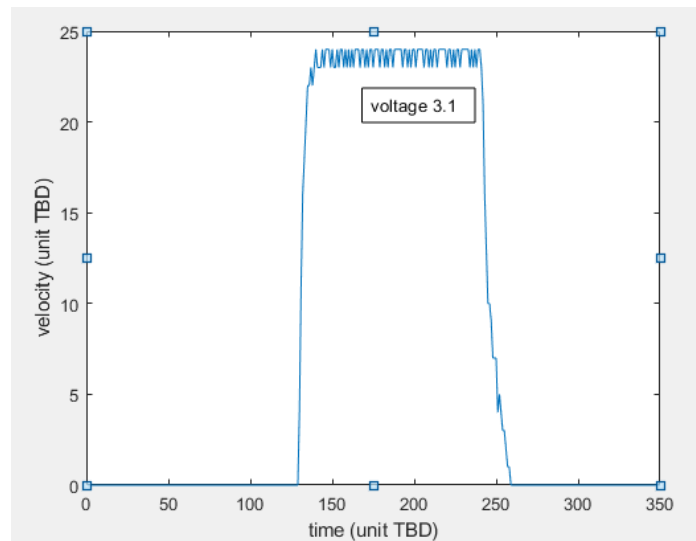


Figure 3: Velocity Step Response Graph

Questions:

- Why is a value of 255 used?

- Using the plot commands above zoom in on part of the velocityuplot that contains the velocity step response similar to the plot above. If you use the plot commands above what would be the units of each axis? Be sure to label them.

Part 2: Motor Logic: Direction and Magnitude

Background Information:

You do not have to make any Simulink diagrams in this section, this section is information only.

To regulate the speed a PWM signal is used to quickly switch the output on and off so that the average output voltage can be controlled. In addition, the correct switching logic needs to be implemented so that the motor can change direction based on the sign of the input.

The subsystem block below is used to correctly output the correct magnitude and logic of the PWM if the only input is a scalar input voltage (which could be negative). **You do not have to make this Simulink diagram – it is shown for reference only.**

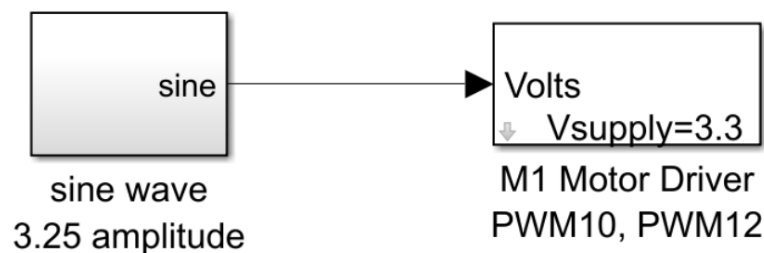


Figure 4: Motor driver block to control magnitude and direction

To correctly use this block, you must specify Vsupply. Vsupply is the maximum high-power voltage the driver is capable of providing. The voltage it can supply is usually the high-power voltage input supplied to the driver minus the losses associated with the driver. It is also depended on the computer power supply and USB type (2.0, 3.0, type C) or battery supply.

For example, if the SN754410 driver is supplied a high-power voltage of 4.5 volts from the USB, the max voltage observed at the output pins might be 3.3 volts (ideally this would be 4.5 volts). Due to driver losses the max voltage the application will receive is 3.3 volts.

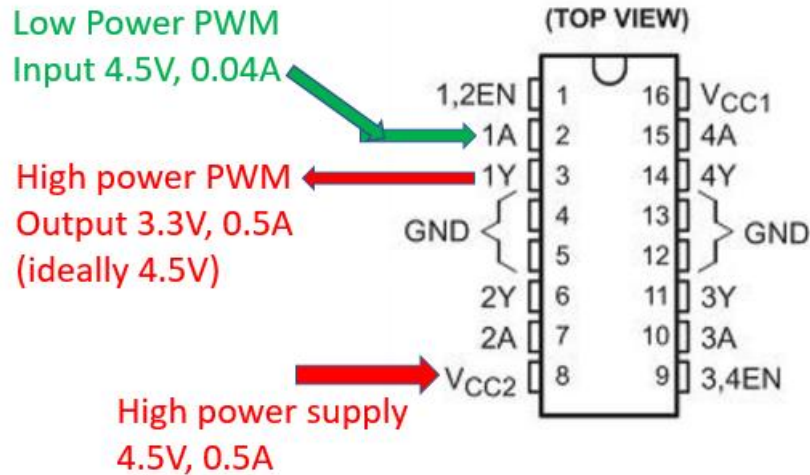


Figure 5: V_{supply} for this example is 3.3 volts - the maximum voltage the driver is capable of supplying. Even though the input high power supply voltage is 4.5volts, due to driver losses the max available at the output is 3.3 volts.

To translate your desired voltage in Simulink to the PWM block you have to scale your voltage by V_{supply} : $255/V_{supply}$, or in this example $255/3.3.e$

For the SN754410 driver this is approximately the voltage supply minus 1.2 volts which are the losses from the driver. For the DRV8833 driver the loss is about 0.2 volts.

The contents of this subsystem are shown below. Each input voltage is scaled from voltage before being sent to the PWM block used in the first part of this lab. Looking inside the “M1 Volts to PWM” for the SN754410 driver to see the logic used to scale and

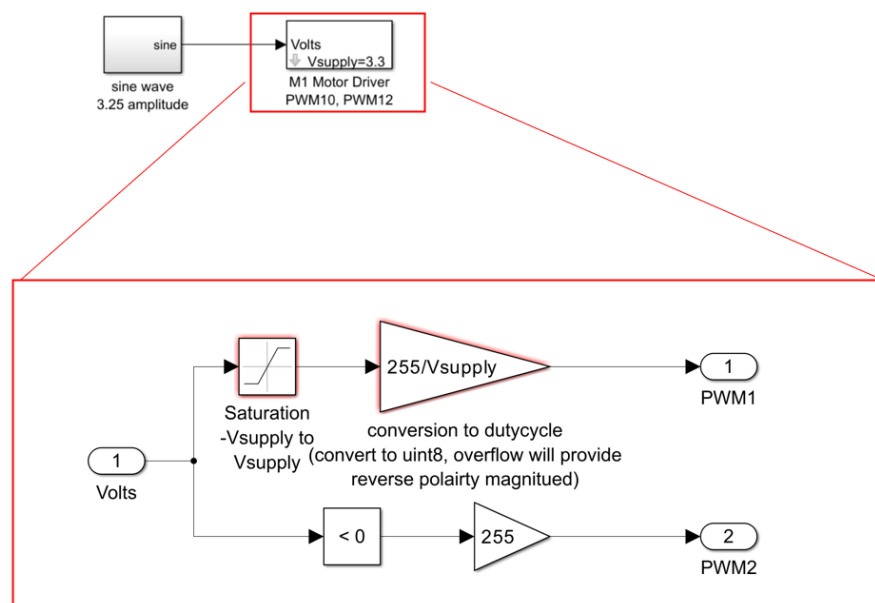


Figure 6: Example Logic for controlling magnitude and direction of motor voltage for the SN754410 driver (for reference only - do not build this diagram, the contents of your driver block may look different)

The logic to control direction and magnitude for the DRV8833 Motor driver:

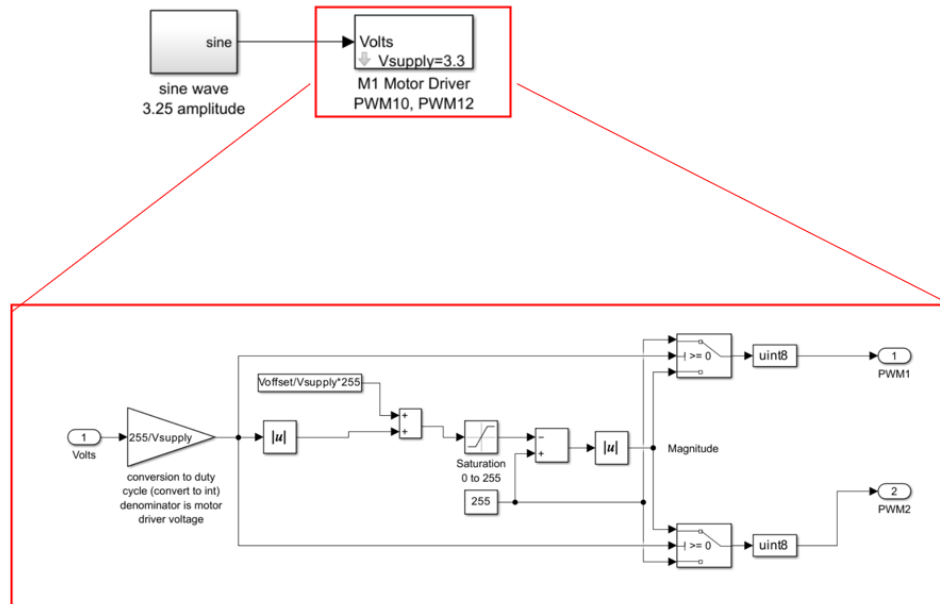


Figure 8: Example Logic for controlling magnitude and direction of motor voltage for the DRV8833 driver (for reference only - do not build this diagram, the contents of your driver block may look different)

Notice that different driver chips might require different logic to obtain the desired voltage response. The documentation for the chip will provide guidance for the different ways it can be used.

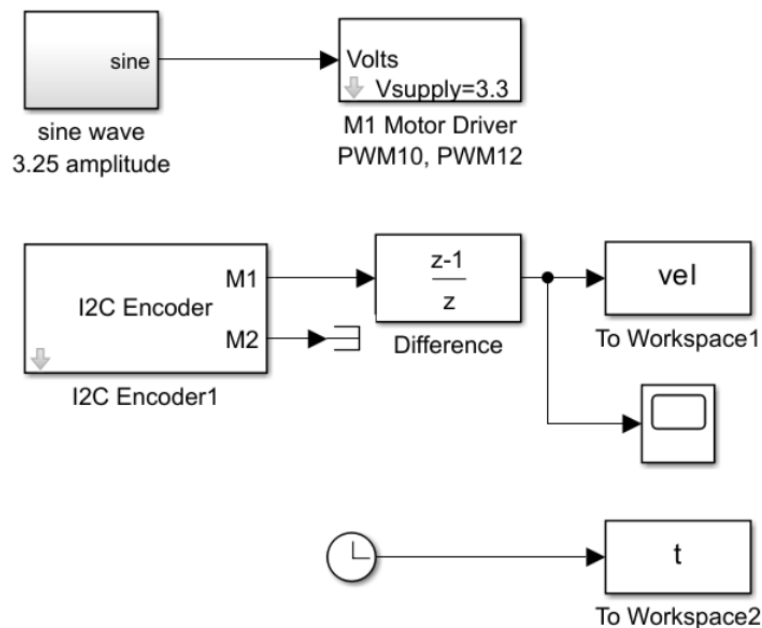
As shown in Figure 7, It performs the 3 following basic functions (you do not have to make the block diagram in Figure 7):

- It scales the input by the voltage that is being supplied to the driver (V_{supply}), this appropriately scales the voltage to PWM (a number from 0-255).
- Based on the sign of the input – it will switch one of the motor control lines to positive voltage, or ground (controlling the direction of the motor)
- Based on the magnitude of the input it controls the magnitude of the resulting PWM

Different motor driver chips will require different logic depending on how you want them to perform.

Simulink Model – Magnitude and Direction Control with Motor Driver Block

Use the motor driver block in the demo file to build the following Simulink diagram in the figure below to generate a sinusoidal motor response.



- Set the Vsupply of the driver block to 3.25 and set the parameters of the sine wave so it has a magnitude of the same volts as Vsupply 3.25 volts and a period of around 3 seconds. **Use the sine wave block from the demo file.**

Questions:

- Provide a single plot vs time of the sinusoidal velocity in external mode with a solver sample time of .03 and at .003 seconds. Do not modify the units of the vel signal. Explain why these two signals look different (look at the values of the magnitude of the response).

Part 3: Using the Discrete Derivative Block

In the previous sections the Difference block was used to calculate velocity. The difference block ensured the resulting number would be integers (the encoder block outputs number of counts) this could allow the data to be sent as an unsigned integer. Using the derivative block will automatically divide by the sample time and the data must be represented as a floating-point number. A unit conversion is then needed to convert from counts to radians/second. **(You will need to use the correct unit conversion for *you* motor and the correct blocks)**

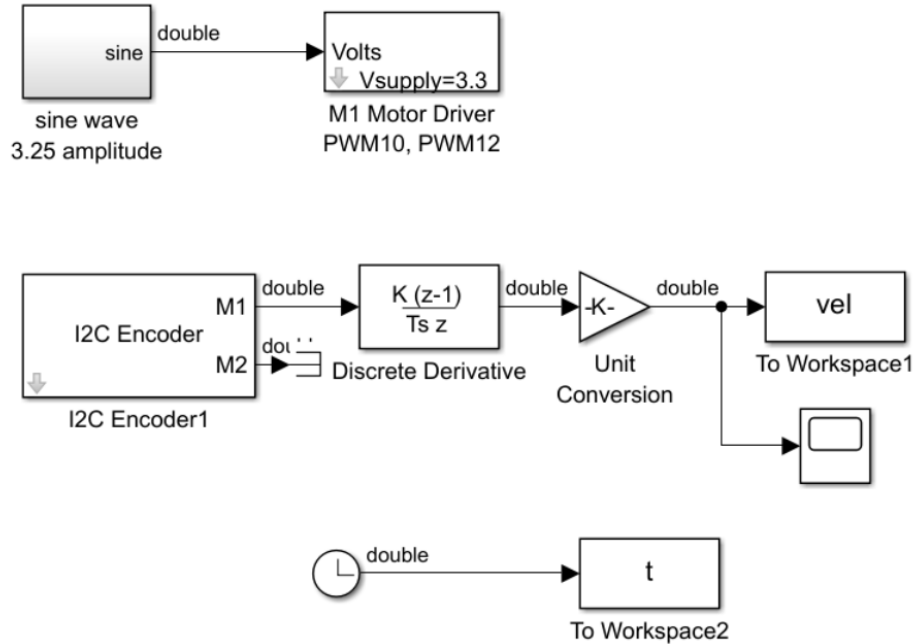


Figure 7: Use the correct blocks for your system

Questions:

Provide a plot of the sinusoidal velocity in external mode at .03 at .003 seconds. Using the derivative block. Label the axes and title appropriately with correct units. Explain what is different than the previous graph and why it is different.